

MEI_JAVA_Week 8_COD

1)

```
try {  
    Class.forName("com.mysql.jdbc.Driver");  
  
    String url = "jdbc:mysql://localhost:3306/ri_db";  
    String username = "test";  
    String password = "test123";  
  
    conn = DriverManager.getConnection(url, username, password);  
  
    System.out.println("Connected to the database successfully!");  
  
}
```

2)

```

-- Insert data into Restaurants
INSERT INTO Restaurants (RestaurantID, RestaurantName, Location, CuisineType)
VALUES
    (1, 'Pasta Paradise', 'New York', 'Italian'),
    (2, 'Sushi World', 'Tokyo', 'Japanese'),
    (3, 'Taco Haven', 'Los Angeles', 'Mexican'),
    (4, 'Curry Corner', 'London', 'Indian'),
    (6, 'Pasta Express', 'Chicago', 'Italian');

-- Insert data into Orders
INSERT INTO Orders (OrderID, RestaurantID, CustomerID, OrderDate, Amount)
VALUES
    (1, 1, 101, '2024-07-15', 45.00),
    (2, 2, 102, '2024-06-20', 55.00),
    (5, 5, 105, '2024-06-30', 25.00),
    (6, 6, 106, '2024-05-15', 60.00),
    (8, 901, 108, '2024-07-01', 20.00);

-- Update data in Restaurants
UPDATE Restaurants
SET Location = 'Rome'
WHERE CuisineType = 'Italian'
    AND Location <> 'Rome';

-- Delete data from Orders
DELETE FROM Orders
WHERE (OrderDate < '2023-12-01' AND Amount > 50)
    OR RestaurantID = 901;

```

3)

```

SELECT AVG(SALE_AMOUNT) AS AVG_SALE_AMOUNT
FROM SALES_TRANSACTIONS;

SELECT SUM(SALE_AMOUNT) AS TOTAL_SALES_AMOUNT
FROM SALES_TRANSACTIONS;

```

4)

```

SELECT
    e.employee_name,
    e.salary,
    d.department_name
FROM
    Employees e
INNER JOIN
    Departments d
ON
    e.department_id = d.department_id;

```

5)

```

SELECT Name,location from Employee
where id in(select employeeid
FROM Project
where Manager='Charles');

```

MEI_JAVA_Week 8_CY

1)

```

class InsertUserRecord {
    public static void insertUser(String username, String email) {
        String url = "jdbc:mysql://localhost:3306/ri_db";
        String dbUser = "test";
        String dbPassword = "test123";

        String sql = "INSERT INTO users (username, email) VALUES (?, ?)";

        try (Connection conn = DriverManager.getConnection(url, dbUser, dbPassword);
            PreparedStatement ps = conn.prepareStatement(sql)) {

            ps.setString(1, username);
            ps.setString(2, email);

            int rowsInserted = ps.executeUpdate();

            if (rowsInserted > 0) {
                System.out.println("Record inserted successfully for " + username + " with email " + email + ".");
            } else {
                System.out.println("Record insertion failed.");
            }
        } catch (SQLException e) {
            System.out.println("Record insertion failed.");
        }
    }
}

```

2)

```
SELECT CUSTOMER_ID, AVG(RATING) AS AVG_RATING
```

```
FROM
```

```
    CUSTOMER_FEEDBACK
```

```
GROUP BY
```

```
    CUSTOMER_ID;
```

```
SELECT CUSTOMER_ID, COUNT(BOOKING_ID) AS TOTAL_BOOKINGS
```

```
FROM
```

```
    BOOKINGS
```

```
GROUP BY
```

```
    CUSTOMER_ID;
```

```
SELECT CUSTOMER_ID, MAX(RATING) AS HIGHEST_RATING
```

```
FROM
```

```
    CUSTOMER_FEEDBACK
```

```
GROUP BY
```

```
    CUSTOMER_ID;
```

```
SELECT CUSTOMER_ID, SUM(NUMBER_OF_TICKETS) AS TOTAL_TICKETS_BOOKED
```

```
FROM
```

```
    BOOKINGS
```

```
GROUP BY
```

```
    CUSTOMER_ID;
```

```
SELECT CUSTOMER_ID, MIN(RATING) AS MIN_RATING
```

```
FROM
```

CUSTOMER_FEEDBACK

GROUP BY

CUSTOMER_ID;

3)

```
SELECT
    P.PROJECT_NAME,
    P.BUDGET,
    E.FIRST_NAME AS LEAD_FIRST_NAME,
    E.LAST_NAME AS LEAD_LAST_NAME
FROM
    PROJECTS P
INNER JOIN
    EMPLOYEES E ON P.TEAM_LEAD_ID = E.EMP_ID;
```

4)

SELECT Name from Employee where id in(select employeeid FROM Project where Project='P4');

5)

```

INSERT INTO Products (ProductID, ProductName, Category, Price, StockQuantity)
VALUES
(1001, 'Smartphone', 'Electronics', 699.99, 5),
(1002, 'Laptop', 'Electronics', 1200.00, 8),
(1003, 'Sofa', 'Furniture', 450.00, 20),
(1004, 'Coffee Maker', 'Home Appliances', 120.00, 12),
(1005, 'Table Lamp', 'Furniture', 85.00, 25);

INSERT INTO Suppliers (SupplierID, SupplierName, ContactEmail)
VALUES
(1301, 'ABC Electronics', 'contact@abcelectronics.com'),
(1302, 'XYZ Furnishings', 'info@xyzfurnishings.com'),
(1303, 'Home Goods Inc', 'support@homegoods.com'),
(1304, 'Tech World', 'service@techworld.com'),
(1305, 'Luxury Living', 'contact@luxuryliving.com');

UPDATE Products
SET Price = Price * 1.10
WHERE Category = 'Electronics'
    AND Price > 500
    AND StockQuantity < 10;

DELETE FROM Products
WHERE (StockQuantity < 5 AND Category <> 'Home Appliances')
    OR ProductID = 1001
    OR ProductID = 1002;

SELECT *
FROM Suppliers
WHERE SupplierID IN (1301, 1302, 1303, 1304);

```

Week-8 - PAH

```

try {
    Class.forName("com.mysql.jdbc.Driver");

    String url = "jdbc:mysql://localhost:3306/ri_db";
    String username = "test";
    String password = "test123";

    conn = DriverManager.getConnection(url, username, password);

    System.out.println("Database connection established!");

}

```

1)

```
SELECT PRICE, COUNT(PRODUCT_ID) AS PRODUCT_COUNT
FROM PRODUCT_PRICES
GROUP BY PRICE
ORDER BY PRICE;
```

```
SELECT PRODUCT_NAME, MIN(PRICE) AS MIN_PRICE
FROM PRODUCT_PRICES
GROUP BY PRODUCT_NAME
ORDER BY MIN_PRICE;
```

```
SELECT PRODUCT_NAME, MAX(PRICE) AS MAX_PRICE
FROM PRODUCT_PRICES
GROUP BY PRODUCT_NAME
ORDER BY MAX_PRICE DESC;
```

```
SELECT PRODUCT_NAME, AVG(PRICE) AS AVG_PRICE
FROM PRODUCT_PRICES
GROUP BY PRODUCT_NAME
ORDER BY AVG_PRICE DESC;
```

2)

```
SELECT
    S1.STORE_NAME,
    S2.STORE_NAME AS PARENT_STORE_NAME
FROM
    STORES S1
INNER JOIN
    STORES S2
ON
    S1.PARENT_STORE_ID = S2.STORE_ID;
```

3)

```
SELECT
    C1.MEMBER_NAME,
    C2.MEMBER_NAME AS SUPERVISOR_NAME
FROM
    CREW_MEMBERS C1
INNER JOIN
    CREW_MEMBERS C2
ON
    C1.SUPERVISOR_ID = C2.MEMBER_ID;
```

4)

```

-- Create Students table
CREATE TABLE Students (
    StudentID INT NOT NULL,
    StudentName VARCHAR(100) NOT NULL,
    Email VARCHAR(100) NOT NULL,
    CONSTRAINT pk_students PRIMARY KEY (StudentID)
);

-- Create ExamResults table with ON DELETE CASCADE and ON UPDATE CASCADE
CREATE TABLE ExamResults (
    ResultID INT NOT NULL,
    StudentID INT NOT NULL,
    Subject VARCHAR(100) NOT NULL,
    Score INT NOT NULL,
    ExamDate DATE NOT NULL,
    CONSTRAINT pk_examresults PRIMARY KEY (ResultID),
    CONSTRAINT fk_examresults_student FOREIGN KEY (StudentID)
        REFERENCES Students(StudentID)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);

```

5)

```

-- Insert records into Students table
INSERT INTO Students (StudentID, StudentName, Email) VALUES
(1, 'Alice Johnson', 'alice@example.com'),
(2, 'Bob Smith', 'bob@example.com'),
(3, 'Charlie Brown', 'charlie@example.com'),
(4, 'David Miller', 'david@example.com'),
(5, 'Emma Watson', 'emma@example.com');

-- Insert records into ExamResults table
INSERT INTO ExamResults (ResultID, StudentID, Subject, Score, ExamDate) VALUES
(101, 1, 'Mathematics', 85, '2024-06-01'),
(102, 2, 'Physics', 78, '2024-06-02'),
(103, 3, 'Chemistry', 90, '2024-06-03'),
(104, 4, 'Biology', 88, '2024-06-04'),
(105, 5, 'Computer Science', 95, '2024-06-05');

-- Delete StudentID 2 (will also delete related ExamResults due to ON DELETE CASCADE)
DELETE FROM Students WHERE StudentID = 2;

-- Insert a new student to allow the update
INSERT INTO Students (StudentID, StudentName, Email)
VALUES (6, 'Frank Wilson', 'frank@example.com');

-- Update StudentID in ExamResults (now it will work due to ON UPDATE CASCADE)
UPDATE ExamResults SET StudentID = 6 WHERE ResultID = 101;

```